

A study on the upstream and downstream manufacturing of Covid-19 vaccine- CoronaVac (Sinovac Life Sciences, Beijing, China)

Coronaviruses are a family of viruses that can cause illnesses such as the common cold, severe acute respiratory syndrome (SARS) and Middle East respiratory syndrome (MERS). In 2019, a new coronavirus was identified as the cause of a disease outbreak that originated in China.



Signs and symptoms of coronavirus disease 2019 (COVID-19) may appear two to 14 days after exposure. This time after exposure and before having symptoms is called the incubation period. You can still spread COVID-19 before you have symptoms, called pre-symptomatic. Common signs and symptoms can include:

- Fever
- Cough
- Tiredness

Early symptoms of COVID-19 may include a loss of taste or smell.

Other symptoms can include:

- Shortness of breath or difficulty breathing
- Muscle aches
- Chills
- Sore throat
- Runny nose
- Headache
- Chest pain

- Pink eye (conjunctivitis)
- Nausea
- Vomiting
- Diarrhea
- Rash

There are several vaccines that are developed against SARSCOV2 virus but manufacturing of only one vaccine is mentioned here.

CoronaVac Vaccine

CoronaVac vaccine falls under the category of inactivated pathogen vaccine.



Initially launched under the name PiCoVacc, CoronaVac, it is a purified, inactivated virus alum-adjuvanted candidate vaccine. The candidate was produced by β -propiolactone-activation of the CN2 strain of SARS-CoV-2 isolated from the bronchoalveolar lavage of a hospitalized patient, this strain is closely related to the 2019-nCoV-BetaCoV Wuhan/WIV04/2019 strain. Inactivated vaccines present some technical challenges as a disadvantage. The inactivation process can sometimes damage the antigens leading to suboptimal immunogenicity. Also, inactivated vaccines commonly need several boost doses to produce strong immune responses and do not traditionally activate cellular responses. More importantly, in order to enhance their capacity to provoke immunity, these vaccines require the addition of adjuvants. Indeed, CoronaVac is an alum-adjuvanted candidate vaccine.

A scientist have described that their candidate was highly immunogenic in BALB/c mice and Wistar rats inducing high levels of anti-S and anti-RBD antibodies that peaked 6 weeks post-immunization but, surprisingly, induced lower anti-Nucleocapsid protein antibody titers. Among the elicited

antibodies, SARS-CoV-2 specific neutralizing antibodies were detected as assessed by microneutralization assays that measure the immunized animal sera capacity to inhibit the SARS-CoV-2 cytopathic effects on Vero cells.

Moreover, immunization of rhesus macaques (*Macaca mulatta*) with CoronaVac and their subsequent exposure to SARS-CoV-2 three weeks post-immunization revealed that the vaccine candidate induced partial or complete protection against the viral challenge as assessed by viral RNA titers in pharynx, crissum, and lung histopathological change.

A Phase 2 randomized, double-blind, placebo-controlled dose-escalation safety and immunogenicity trial was conducted enrolling 600 healthy volunteers between 18 and 59 years old who received two different dosages of the vaccine (3 or 6 µg/0.5 ml) or placebo. CoronaVac was well tolerated at both dosages and most of the adverse reactions were mild. Pain at the injection site was the most common of reported symptoms. CoronaVac dosages induced a seroconversion on more than 90% of the individuals immunized, however, no T cell responses were reported.

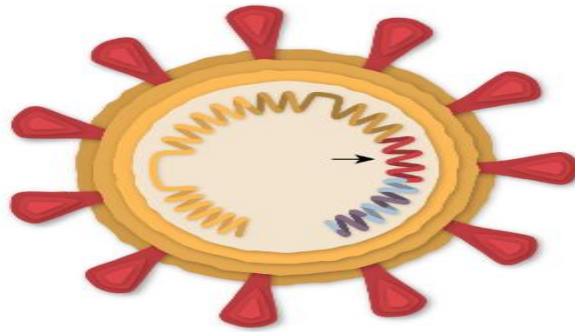
CoronaVac is currently in Phase 3 clinical trials in a two-dose injection regimen with a 14-days interval. These trials will enrol 8870 participants from Brazil), volunteers from Indonesia and Turkey evaluating the efficacy of the vaccine in preventing from COVID-19 and the frequency of adverse effects.

CoronaVac reportedly received an emergency approval for limited use in July in China, whilst Sinovac is scaling up production of their candidate vaccine with an aim to supply Indonesia with 40 million doses by March 2021 and begin worldwide distribution in early 2021

How does Sinovac-CoronaVac vaccine works?

The CoronaVac vaccine works on a similar principle to the Sinopharm, by teaching the immune system to generate antibodies to fight the SARS-CoV-2 and attaching these antibodies to the spike proteins of the virus studded on its surface. While the researchers gathered the virus samples from patients from Italy, Britain, Spain and Switzerland, the one that was used to design the vaccine came from China itself. The virus was then grown in large numbers in Monkey Kidney cells before they were doused with a chemical called beta-propiolactone, the compound that disabled the coronavirus and stopped its replication. However, the spike proteins remained intact and scientists had to mix the inactivated virus to adjuvant, an aluminium-based compound that stimulates the immune system to trigger and boost response to the vaccine. The vaccine once injected into the body is consumed by a type of immune cell called an antigen-presenting cell that rips the inactivated virus displaying its fragments on the surface. According to a report in the New York Times, "helper T cell may detect the fragment. If the fragment fits into one of its surface proteins, the T cell becomes activated and can help recruit other immune cells to respond." Meanwhile, B cells, another type of immune cells, pulls the fragments activating themselves, pouring out antibodies.

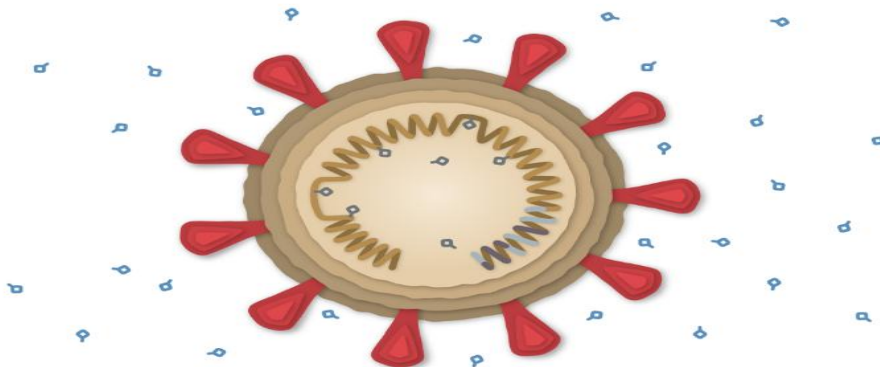
A Vaccine Made From Coronaviruses



CoronaVac works by teaching the immune system to make antibodies against the SARS-CoV-2 coronavirus. The antibodies attach to viral proteins, such as the so-called spike proteins that stud its surface.

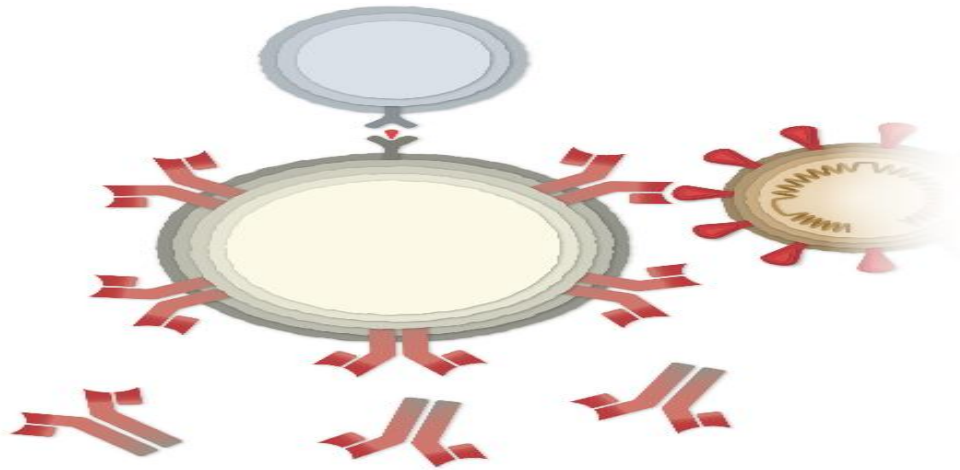
To create CoronaVac, the Sinovac researchers started by obtaining samples of the coronavirus from patients in China, Britain, Italy, Spain and Switzerland. One sample from China eventually served as the basis for the vaccine

Killing the Virus



The researchers grew large stocks of the coronavirus in monkey kidney cells. Then they doused the viruses with a chemical called beta-propiolactone. The compound disabled the coronaviruses by bonding to their genes. The inactivated coronaviruses could no longer replicate. But their proteins, including spike, remained intact. The researchers then drew off the inactivated viruses and mixed them with a tiny amount of an aluminum-based compound called an adjuvant. Adjuvants stimulate the immune system to boost its response to a vaccine.

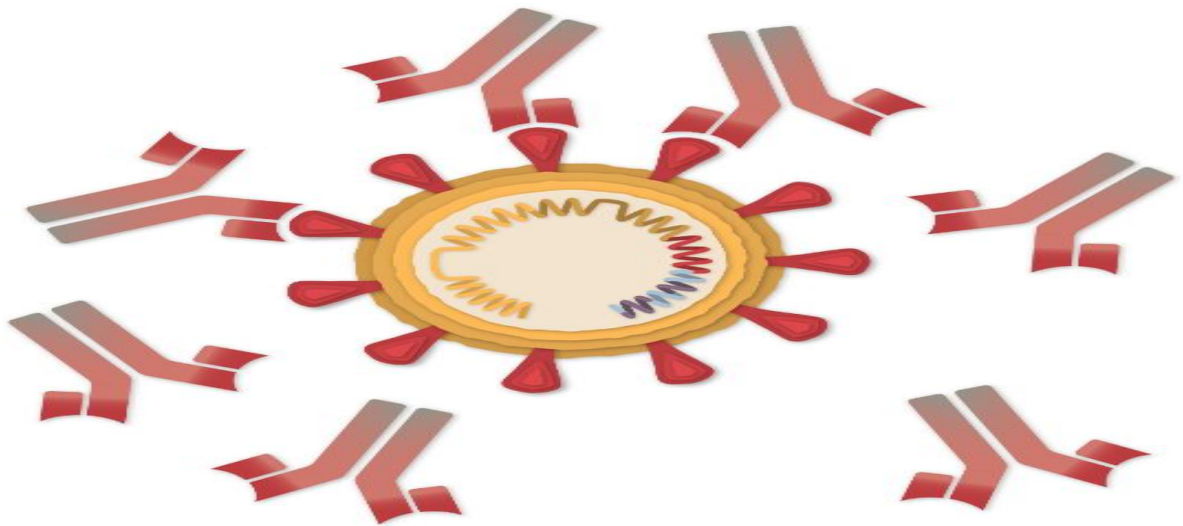
Prompting an Immune Response



Because the coronaviruses in CoronaVac are dead, they can be injected into the arm without causing Covid-19. Once inside the body, some of the inactivated viruses are swallowed up by a type of immune cell called an antigen-presenting cell.

The antigen-presenting cell tears the coronavirus apart and displays some of its fragments on its surface. A so-called helper T cell may detect the fragment. If the fragment fits into one of its surface proteins, the T cell becomes activated and can help recruit other immune cells to respond to the vaccine.

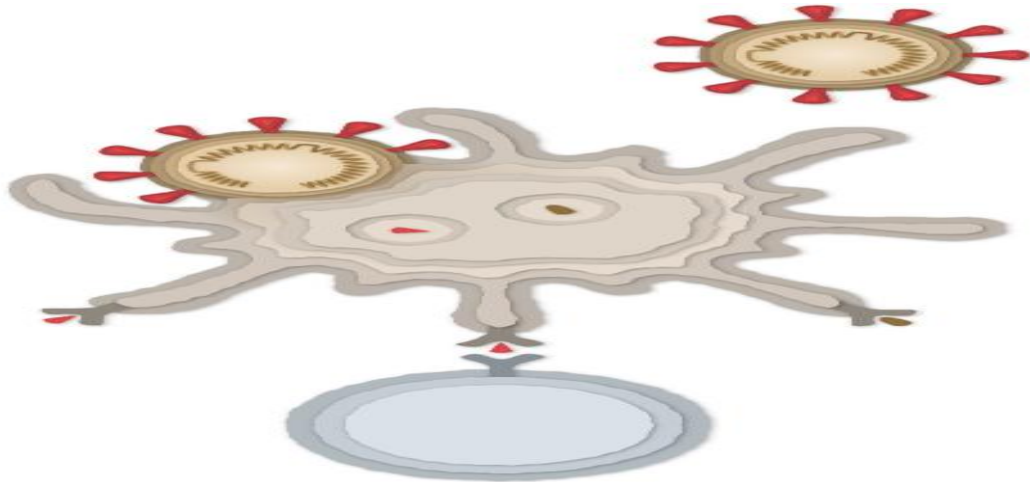
Making Antibodies



Another type of immune cell, called a B cell, may also encounter the inactivated coronavirus. B cells have surface proteins in a huge variety of shapes, and a few might have the right shape to latch onto the coronavirus. When a B cell locks on, it can pull part or all of the virus inside and present coronavirus fragments on its surface.

A helper T cell activated against the coronavirus can latch onto the same fragment. When that happens, the B cell gets activated, too. It proliferates and pours out antibodies that have the same shape as their surface proteins.

Stopping the Virus



Once vaccinated with CoronaVac, the immune system can respond to an infection of live coronaviruses. B cells produce antibodies that stick to the invaders. Antibodies that target the spike protein can prevent the virus from entering cells. Other kinds of antibodies may block the virus by other means.

Remembering the Virus

While CoronaVac can offer some protection against Covid-19, no one can yet say how long that protection lasts. It's possible that the level of antibodies drops over the course of months. But the immune system also contains special cells called memory B cells that might retain information about the coronavirus for years or even decades.

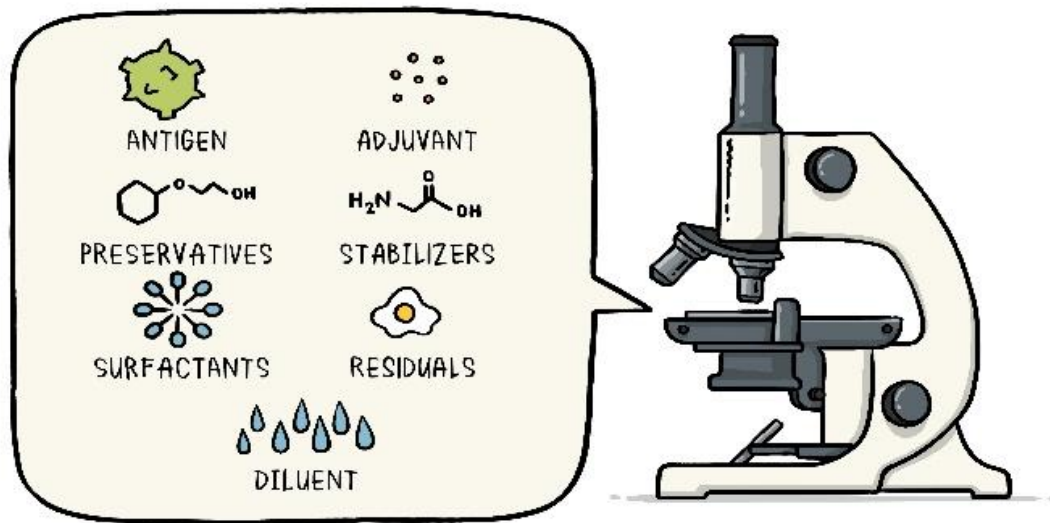
Developing the vaccine

Work on developing the vaccine began in the Sinovac pharmaceutical company in January 2020 and the phase I, II trials were conducted in June 2020. The trials found no adverse effect on the samples. 600 healthy adults aged 18-59 years were randomly assigned to receive two injections of the trial vaccine in the first two phases. In a paper published in medRxiv, the researchers said that CoronaVac was well tolerated, and no dose-related safety concerns were observed. Most of the adverse reactions were mild and the most frequently reported symptoms were pain at the injection site.

What are the ingredients in these vaccines?

Antigen

All vaccines contain an active component (the antigen) which generates an immune response, or the blueprint for making the active component. The antigen may be a small part of the disease-causing organism, like a protein or sugar, or it may be the whole organism in a weakened or inactive form.



Preservatives

Preservatives prevent the vaccine from becoming contaminated once the vial has been opened, if it will be used for vaccinating more than one person. Some vaccines don't have preservatives because they are stored in one-dose vials and are discarded after the single dose is administered. The most commonly used preservative is 2-phenoxyethanol. It has been used for many years in a number of vaccines, is used in a range of baby care products and is safe for use in vaccines, as it has little toxicity in humans.

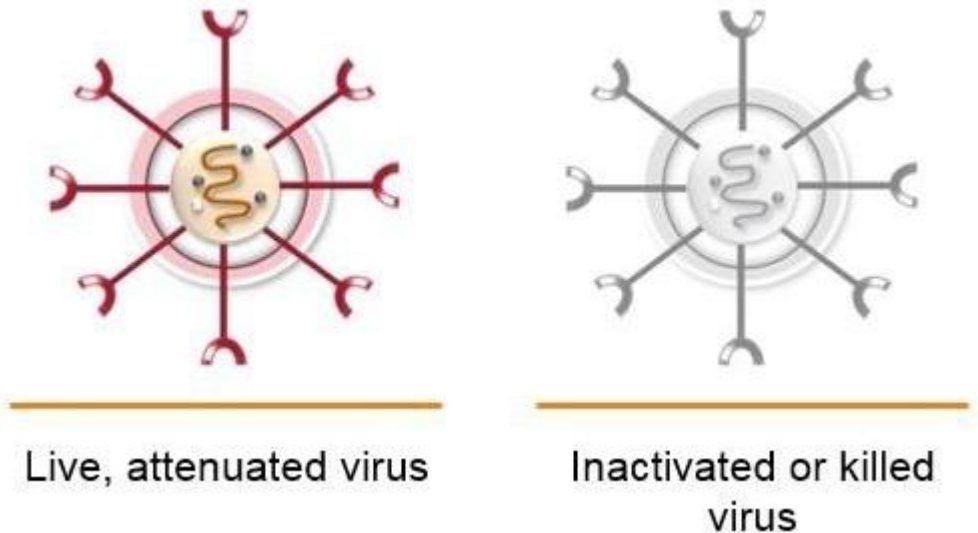
Stabilizers

Stabilizers prevent chemical reactions from occurring within the vaccine and keep the vaccine components from sticking to the vaccine vial.

Stabilizers can be sugars (lactose, sucrose), amino acids (glycine), gelatin, and proteins (recombinant human albumin, derived from yeast).

Surfactants

Surfactants keep all the ingredients in the vaccine blended together. They prevent settling and clumping of elements that are in the liquid form of the vaccine. They are also often used in foods like ice cream.



Residuals

Residuals are tiny amounts of various substances used during manufacturing or production of vaccines that are not active ingredients in the completed vaccine. Substances will vary depending on the manufacturing process used and may include egg proteins, yeast or antibiotics. Residual traces of these substances which may be present in a vaccine are in such small quantities that they need to be measured as parts per million or parts per billion.

Diluent

A diluent is a liquid used to dilute a vaccine to the correct concentration immediately prior to use. The most commonly used diluent is sterile water.

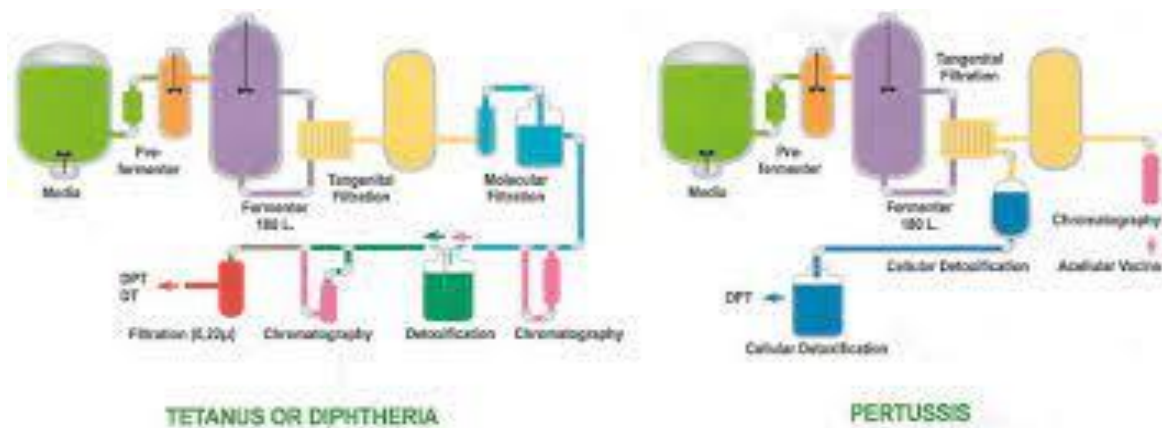
Adjuvant

Some vaccines also contain adjuvants. An adjuvant improves the immune response to the vaccine, sometimes by keeping the vaccine at the injection site for a little longer or by stimulating local immune cells.

The adjuvant may be a tiny amount of aluminium salts (like aluminium phosphate, aluminium hydroxide or potassium aluminium sulphate). Aluminium has been shown not to cause any long-term health problems, and humans ingest aluminium regularly through eating and drinking.

CoronaVac is an inactivated vaccine candidate against COVID-19, created from African green monkey kidney cells (Vero cells) that have been inoculated with SARS-CoV-2 (CN02 strain). At the end of the incubation period, the virus was harvested, inactivated with β -propiolactone, concentrated, purified, and finally absorbed onto aluminium hydroxide. The aluminium hydroxide complex was then diluted in a sodium chloride, phosphate-buffered saline, and water solution before being sterilised and filtered ready for injection.

Stages of vaccine production



To make an inactivated vaccine, the virus has to be grown in culture media. This can often be an initial limiting step in vaccine production, as producers need to understand what conditions encourage viral growth.

Inactivation of the virus is done with heat. Occasionally, inactivation is done with chemicals such as formalin. When the vaccine under production is fractional, meaning it is protein or polysaccharide-based, the vaccine undergoes further purification so that only the subunits of interest remain. Sinovac Biotech has created an inactivated whole virus particle vaccine, with an extra immune booster referred to as alum. This has been successful against SARS viruses. The Sinovac vaccine has been approved for use in China with emergency use in some other countries. Various other inactivated COVID-19 vaccines are in development, with three others at various levels of approval.

The primary purpose of virus cultivation is:

1. To isolate and identify viruses in clinical samples.
2. To do research on viral structure, replication, genetics and effects on host cell.
3. To prepare viruses for vaccine production.

Cultivation of viruses can be discussed under following headings:

1. Animal Inoculation
2. Inoculation into embryonated egg
3. Cell Culture

Inactivated vaccines could be advantageous in the fight against COVID-19 because they can be easily scaled up to large-scale production in many countries. However, the potential effectiveness of inactivated vaccines against COVID-19 is unknown. Current trials with an inactivated vaccine, where

inactivation was done with formaldehyde and ultraviolet light, have been effectively used in a mouse model.

Vaccine production has several stages. Process of vaccine manufacture has the following steps:

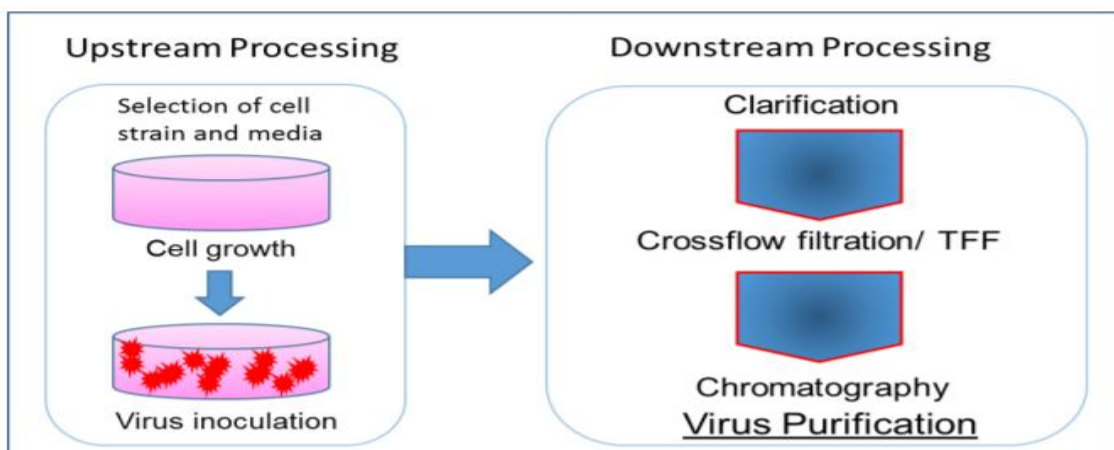
- Inactivation – This involves making of the antigen preparation
- Purification – The isolated antigen is purified
- Formulation – The purified antigen is combined with adjuvants, stabilizers and preservatives to form the final vaccine preparation.

Generating the antigen from the microbe

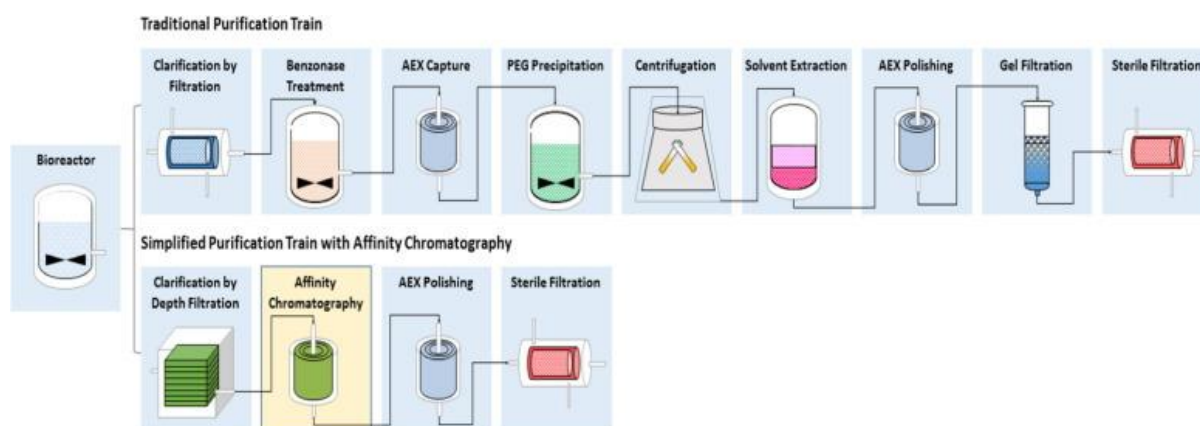
The initial production involves generation of the antigen from the microbe. For this the virus or microbe is grown either on primary cells such as chicken eggs or on cell lines or cultured human cells. Bacteria against which the vaccines are developed may be grown in bioreactors. The antigen may also be a toxin or toxoid from the organism or it may be part of the microorganism as well. Proteins or parts from the organism can be generated in yeast, bacteria, or cell cultures. Viruses may be weakened using chemicals or heat to make the vaccine.

Isolation of the antigens

After the antigen is generated, it is isolated from the cells used to generate it. For weakened or attenuated viruses no further purification may be required. Recombinant proteins need many operations involving ultrafiltration and column chromatography for purification before they are ready for administration.



Recovery and purification process of these vaccines



After harvesting viruses from the host cells, downstream processing deals with the purification and concentration of the viral products. Purification is aimed at removing contaminating host cell proteins, DNA, and impurities introduced during the purification process. Using a variety of physical properties as size, isoelectric potential (pI), and the hydrophobicity different methods are selected for purification. There are several methods for virus purification. Commonly used methods for virus purification are ultrafiltration and chromatography. Membrane-based ultrafiltration is typically used for partial purification and chromatography is usually used for final product purification. Other purification methods are depending on the viral size and properties such as ultracentrifugation which uses density based separation, precipitation by chemicals, PEG, flocculation using salts, and membrane absorption which uses charge based properties of viruses.

Ultracentrifugation

One of the methods employed for virus purification is ultracentrifugation. Density gradient centrifugation using cesium chloride or iodixanol gradient is one of the well known and established method for the virus in preclinical studies. Also, there are concerns over CsCl toxicity and its complete removal after viral purification can be challenging. While ultracentrifugation leads to very pure products, the lack of scale-up options makes it undesirable for manufacturing. This method can be used for limited sample separation by using ultracentrifugation, which uses very high rotational speeds.

Chromatography

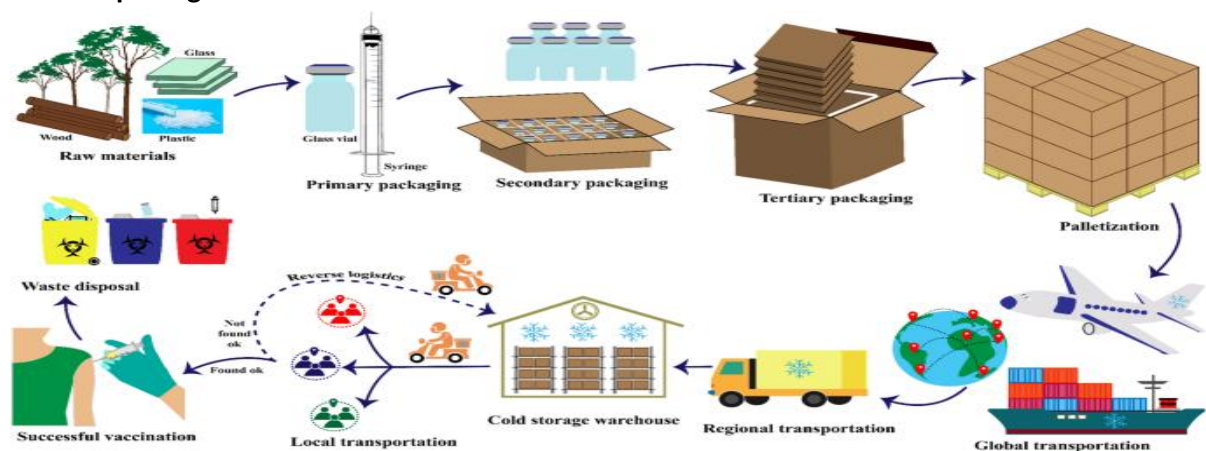
In the biopharmaceutical industry, chromatography is the most commonly used separation and purification technology due to its easy applications to all products and its high resolution. Chromatography separation is based on the different interactions between viruses and the surface of chromatographic beads such as size based (size exclusion chromatography), charge based (ion exchange chromatography), and hydrophobicity based (hydrophobic interaction chromatography HIC, reverse phase chromatography). Ion exchange chromatography is charge based separation depending on the interactions between charged particles and ionic ligands on the chromatography

beads. Therapeutic molecules bind to the column under low salt condition and elution is done using high salt gradient because the ionic interaction is disrupted by the high ionic strength solution.

Precipitation and Flocculation

Precipitation or flocculation is typically followed by centrifugation or filtration to either remove or recover the flocculated/precipitated species. Flocculation prior to filtration is highly desirable as it can reduce the membrane fouling, increase product recovery and decrease the number of steps for final product purification

How it's packaged



Once the vaccine has been made in bulk quantities, it is bottled in glass vials and then carefully packaged for safe cold storage and transport.

Vaccine packaging must be able to withstand extreme temperatures, as well as the risks involved in being transported globally. Therefore, vaccine vials are most commonly made from glass, as it is durable and able to maintain its integrity in extreme temperatures.

How it's stored

When a vaccine is too hot or too cold, it becomes less effective or even inactive. If stored at the incorrect temperature, vaccines can be ruined or unsafe for use. Most vaccines require refrigerated storage at between 2 and 8 °C. Some vaccines require temperatures as cold as -20°C. Some of the newer vaccines need to be kept ultra cold at -70°C. For frozen vaccines some of them can be safely stored for a limited time between 2 and 8°C.

Regular refrigerators cannot maintain an even temperature consistently, so specialized medical refrigerators are required for these precious products.

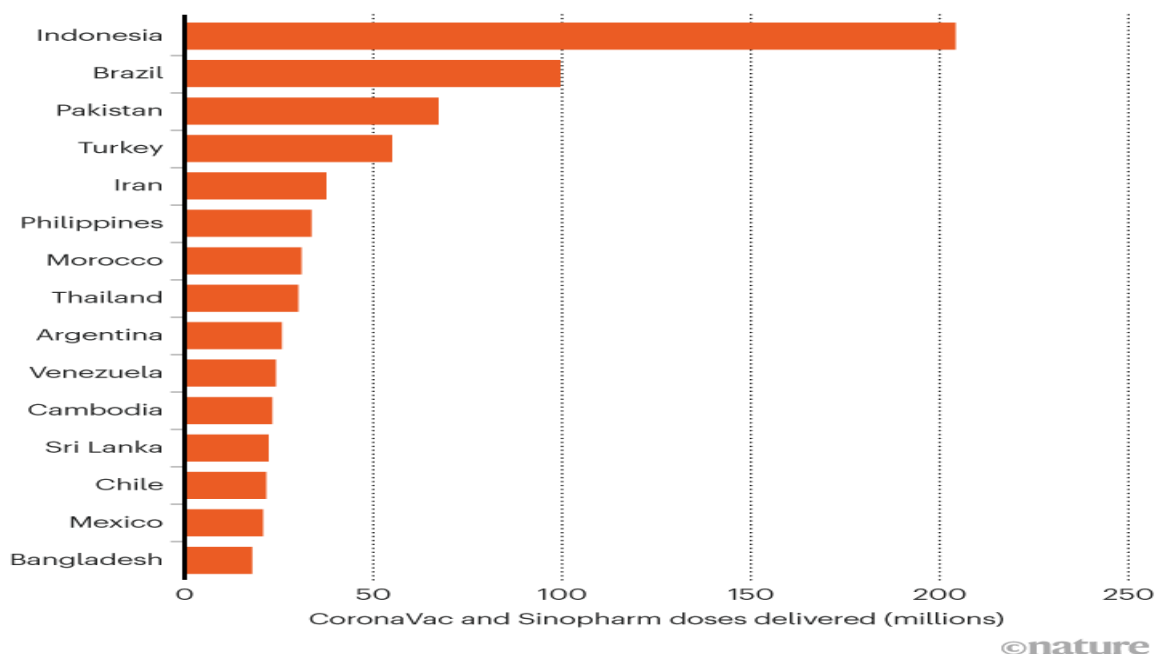
How it's shipped

To maintain this cold chain, vaccines are shipped using specialized equipment that does not compromise the integrity of the product. Once shipments land in the destination country, refrigerated lorries transport the vaccines from the airport to the warehouse cold room. From there, portable iceboxes are used to transport vaccines from the cold room to regional centres where they're stored in refrigerators. If vaccination takes place outside of the regional facility, the final step often requires portable iceboxes to transport the goods to local areas for vaccination campaigns. New technologies have invented some portable devices that can keep vaccines at their cold temperature for several days without needing electricity.

Discussion

BIGGEST TAKERS FOR CHINA'S VACCINES

More than two billion doses of China's CoronaVac and Sinopharm vaccines have been administered within China, but nearly a billion doses have gone to 110 other nations.



According to the WHO, vaccination is a simple, safe and effective way to protect against harmful diseases before coming into contact with them, as it activates the body's natural defenses to learn to resist specific infections and strengthen the immune system.

In this sense, vaccination against COVID-19 will reduce the risk of becoming seriously ill and dying, since the person will be better protected. Immunity will not be 100%, since a vaccinated person can still catch the disease; however, the consequences for the body are expected to be much less.

The main benefits are:

- COVID-19 vaccines can also prevent you from becoming seriously ill even if you contract the virus.

- TAll Covid-19 vaccines are safe and effective.
- By getting vaccinated yourself, you also protect the people around you.
- It is a safer way to develop immunity.

China's CoronaVac and Sinopharm vaccines account for almost half of the 7.3 billion COVID-19 vaccine doses delivered globally, and have been enormously important in fighting the pandemic, particularly in less wealthy nations. Research by Barral-Netto and his colleagues found that CoronaVac prevented only 33% of deaths due to COVID-19 in people 90 and older. Neither study has been peer-reviewed, but Villela says they influenced Brazil's government to start giving people older than 70 a third shot of an mRNA or viral-vector vaccine in August — that decision has now been extended to people older than 60. The inactivated vaccine method is a much more well-established way of designing a vaccine. Inactivated vaccines are typically easy to manufacture at large scale and have an excellent safety record. However, they tend to produce a weaker immune response than vaccines that use other designs.

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